P2 Chapter 9: Differentiation

Differentiating Exponents and Logs

Differentiation exponential and log functions

$$\frac{d}{dx}(e^x) = \boldsymbol{e^x}$$

We know from Year 1 how to differentiate e^x. Can you guess what happens with e^{kx}?

$$\frac{d}{dx}\left(e^{kx}\right) = \boldsymbol{ke^{kx}}$$

$$\frac{d}{dx}(\sin kx) = \boldsymbol{k\cos kx}$$

$$\frac{d}{dx}(\ln x) = \boldsymbol{\frac{1}{x}}$$

$$\frac{d}{dx}(\ln kx) = \boldsymbol{\frac{d}{dx}(\ln k + \ln x) = \frac{1}{x}}$$

i.e. Scaling the input of a log does not affect the gradient. This variant occurs regularly in exams!

$$\frac{d}{dx}(a^x) = \boldsymbol{a^x \ln a}$$

i.e. When we differentiate an exponential function, we multiply by ln of the base. We will prove this after we cover implicit differentiation.

$$\frac{d}{dx}\left(a^{kx}\right) = \boldsymbol{a^{kx} k \ln a}$$

Quickfire Questions

$\frac{d}{dx}(3^x) =$?

$\frac{d}{dx}(x^3) =$?

$\frac{d}{dx}(\ln(3x)) =$?

$\frac{d}{dx}(3^{2x}) =$?

$\frac{d}{dx}(2^{3x}) =$?

$\frac{d}{dx}(5\ln x) =$?

$\frac{d}{dx}\left(e^{\frac{1}{2}x}\right) =$?

$\frac{d}{dx}(5\ln(2x)) =$?

$\frac{d}{dt}(9^t) =$?

$\frac{d}{dx}\left(5(4^x)\right) =$?

$\frac{d}{dx}(x^4) =$?

$\frac{d}{dx}(\ln 6x) =$?

$\frac{d}{dx}(6\ln x) =$?

$\frac{d}{dx}(3e^{2x}) =$?

$\frac{d}{dx}(e^{-x}) =$?

'Meatier' Example:

~~bun~~ pun intended

A rabbit population P after t years can be modelled using $P = 1000(2^t)$. Determine after how many years the rate of population increase will reach 20,000 rabbits per year.

?

Quickfire Questions

$$\frac{d}{dx}(3^x) = \mathbf{3^x \ln 3}$$
$$\frac{d}{dx}(x^3) = \mathbf{3x^2}$$
$$\frac{d}{dx}(\ln(3x)) = \mathbf{\frac{1}{x}}$$
$$\frac{d}{dx}(3^{2x}) = \mathbf{3^{2x}(2 \ln 3)}$$
$$\frac{d}{dx}(2^{3x}) = \mathbf{2^{3x}(3 \ln 2)}$$
$$\frac{d}{dx}(5 \ln x) = \mathbf{\frac{5}{x}}$$
$$\frac{d}{dx}\left(e^{\frac{1}{2}x}\right) = \mathbf{\frac{1}{2}e^{\frac{1}{2}x}}$$
$$\frac{d}{dx}(5 \ln(2x)) = \mathbf{\frac{5}{x}}$$
$$\frac{d}{dt}(9^t) = \mathbf{9^t \ln 9}$$
$$\frac{d}{dx}\left(5(4^x)\right) = \mathbf{5(4^x) \ln 4}$$
$$\frac{d}{dx}(x^4) = \mathbf{4x^3}$$

$$\frac{d}{dx}(\ln 6x) = \mathbf{\frac{1}{x}}$$
$$\frac{d}{dx}(6 \ln x) = \mathbf{\frac{6}{x}}$$
$$\frac{d}{dx}(3e^{2x}) = \mathbf{6e^{2x}}$$
$$\frac{d}{dx}(e^{-x}) = \mathbf{-e^{-x}}$$

'Meatier' Example:

~~bun~~ pun intended

A rabbit population P after t years can be modelled using $P = 1000(2^t)$. Determine after how many years the rate of population increase will reach 20,000 rabbits per year.

$$\mathbf{\frac{dP}{dt} = 1000(2^t) \ln 2 = 20\,000}$$
$$\mathbf{2^t = \frac{20000}{1000 \ln 2} = \frac{20}{\ln 2}}$$
$$\mathbf{t = \log_2\left(\frac{20}{\ln 2}\right) = 4.85\ years\ (3sf)}$$

Test Your Understanding

Differentiate $y = (e^x + 2)^2$ *(Hint: Expand first)*

?

This one is a bit of a headscratcher (pun intended)

A child has headlice and his parents treat it using a special shampoo. The population P of headlice after t days can be modelled using $P = 460(3^{-2t})$

a) Determine how many days have elapsed before the child has 20 headlice left.

b) Determine the rate of change of headlice after 3 days.

a

?

b

?

Test Your Understanding

Differentiate $y = (e^x + 2)^2$ *(Hint: Expand first)*

$$y = e^{2x} + 4e^x + 4$$
$$\frac{dy}{dx} = 2e^{2x} + 4e^x$$

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A child has headlice and his parents treat it using a special shampoo. The population P of headlice after t days can be modelled using $P = 460(3^{-2t})$

a) Determine how many days have elapsed before the child has 20 headlice left.

b) Determine the rate of change of headlice after 3 days.

a

$$20 = 460(3^{-2t})$$
$$3^{-2t} = \frac{1}{23}$$
$$-2t = \log_3\left(\frac{1}{23}\right) = -2.85\ldots$$
$$t = 1.43\ days$$

b

$$\frac{dP}{dt} = -920(3^{-2t})\ln 3$$

When $t = 3$,

$$\frac{dP}{dt} = -920(3^{-6})\ln 3$$
$$= -1.39\ headlice/day$$

Exercise 9.2

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Homework Exercise

1 **a** Find $\frac{dy}{dx}$ for each of the following:

a $y = 4e^{7x}$ **b** $y = 3^x$ **c** $y = \left(\frac{1}{2}\right)^x$ **d** $y = \ln 5x$

e $y = 4\left(\frac{1}{3}\right)^x$ **f** $y = \ln(2x^3)$ **g** $y = e^{3x} - e^{-3x}$ **h** $y = \frac{(1 + e^x)^2}{e^x}$

2 Find $f'(x)$ given that:

a $f(x) = 3^{4x}$ **b** $f(x) = \left(\frac{3}{2}\right)^{2x}$ **c** $f(x) = 2^{4x} + 4^{2x}$ **d** $f(x) = \frac{2^{7x} + 8^x}{4^{2x}}$

Hint In parts **c** and **d**, rewrite the terms so that they all have the same base and hence can be simplified.

3 Find the gradient of the curve $y = (e^{2x} - e^{-2x})^2$ at the point where $x = \ln 3$.

4 Find the equation of the tangent to the curve $y = 2^x + 2^{-x}$ at the point $\left(2, \frac{17}{4}\right)$. **(6 marks)**

5 A curve has the equation $y = e^{2x} - \ln x$. Show that the equation of the tangent at the point with x-coordinate 1 is

$$y = (2e^2 - 1)x - e^2 + 1$$

(6 marks)

Homework Exercise

5 A curve has the equation $y = e^{2x} - \ln x$. Show that the equation of the tangent at the point with x-coordinate 1 is

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(6 marks)

6 A particular radioactive isotope has an activity, R millicuries at time t days, given by the equation $R = 200 \times 0.9^t$. Find the value of $\frac{dR}{dt}$, when $t = 8$.

7 The population of Cambridge was 37 000 in 1900, and was about 109 000 in 2000. Given that the population, P, at a time t years after 1900 can be modelled using the equation $P = P_0 k^t$,

a find the values of P_0 and k

b evaluate $\frac{dP}{dt}$ in the year 2000

c interpret your answer to part **b** in the context of the model.

8 A student is attempting to differentiate $\ln kx$. The student writes:

$$y = \ln kx, \text{ so } \frac{dy}{dx} = k \ln kx$$

Explain the mistake made by the student and state the correct derivative.

9 Prove that the derivative of a^{kx} is $a^{kx} k \ln a$. You may assume that the derivative of e^{kx} is ke^{kx}.

(4 marks)

Homework Exercise

10 $f(x) = e^{2x} - \ln(x^2) + 4,\ x > 0$

a Find $f'(x)$. **(3 marks)**

The curve with equation $y = f(x)$ has a gradient of 2 at point P. The x-coordinate of P is a.

b Show that $a(e^{2a} - 1) = 1$. **(2 marks)**

11 A curve C has equation

$$y = 5\sin 3x + 2\cos 3x,\ -\pi \leqslant x \leqslant \pi$$

a Show that the point P (0, 2) lies on C. **(1 mark)**

b Find an equation of the normal to the curve C at P. **(5 marks)**

12 The point P lies on the curve with equation $y = 2(3^{4x})$. The x-coordinate of P is 1.
Find an equation of the normal to the curve at the point P in the form $y = ax + b$, where a and b are constants to be found in exact form. **(5 marks)**

Challenge

A curve C has the equation $y = e^{4x} - 5x$. Find the equation of the tangent to C that is parallel to the line $y = 3x + 4$.

Homework Answers

1 **a** $28e^{7x}$ **b** $3^x \ln 3$ **c** $\left(\frac{1}{2}\right)^x \ln\frac{1}{2}$ **d** $\frac{1}{x}$

e $4\left(\frac{1}{3}\right)^x \ln\frac{1}{3}$ **f** $\frac{3}{x}$ **g** $3e^{3x} + 3e^{-3x}$ **h** $-e^{-x} + e^x$

2 **a** $3^{4x} 4 \ln 3$ **b** $\left(\frac{3}{2}\right)^{2x} 2 \ln\frac{3}{2}$

c $2^{4x} 8 \ln 2$ **d** $2^{3x} 3 \ln 2 - 2^{-x} \ln 2$

3 323.95 **4** $4y = 15 \ln 2(x-2) + 17$

5 $\frac{dy}{dx} = 2e^{2x} - \frac{1}{x}$ At $x = 1$, $y = e^2$, $\frac{dy}{dx} = 2e^2 - 1$

Equation of tangent: $y - e^2 = (2e^2 - 1)(x - 1)$

$\Rightarrow y = (2e^2 - 1)x - 2e^2 + 1 + e^2 \Rightarrow y = (2e^2 - 1)x - e^2 + 1$

6 −9.07 millicuries/day

7 **a** $P_0 = 37\,000$, $k = 1.01$ (2 d.p.) **b** 1178

c The rate of change of the population in the year 2000

8 The student has treated "ln kx" as if it is "e^{kx}" – they have applied the incorrect standard differential.

Correct differential is: $\frac{1}{x}$

9 Let $y = a^{kx} \Rightarrow y = e^{\ln a^{kx}} = e^{kx \ln a}$

$\frac{dy}{dx} = k \ln a \, e^{kx \ln a} = k \ln a \, e^{\ln a^{kx}} = a^{kx} k \ln a$

10 **a** $2e^{2x} - \frac{2}{x}$

b $2e^{2a} - \frac{2}{a} = 2 \Rightarrow 2ae^{2a} - 2 = 2a \Rightarrow a(e^{2a} - 1) = 1$

11 **a** $5 \sin(3 \times 0) + 2 \cos(3 \times 0) = 0 + 2 = 2 = y$

When $x = 0$, $y = 2$, therefore (0, 2) lies on C.

b $y = -\frac{1}{15}x + 2$

12 $y = -\frac{1}{648 \ln 3}x + \frac{1}{648 \ln 3} + 162$

Challenge

$y = 3x - 2 \ln 2 + 2$